OUTCOME-BASED EDUCATION (OBE) SYSTEM

Complete Technical and Academic Reference Manual

Comprehensive Reference for Deans, HoDs, QEC Auditors, Assessment Specialists & Engineers

Compliance: Pakistan HEC OBE Framework / Washington Accord / NCEAC Computing Accreditation

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Section Identifier	Core Academic Domain Covered
Part A: System Overview	Educational rationale, institutional roles, Washington Accord criteria, and accreditation scope.
Part B: Codebase Architecture	Modular React 19 architecture, dataStore state machine, and Express API routing.
Part C: Database Design & Schema	PostgreSQL entity relationships, table schemas, foreign key cascade constraints, and RLS.
Part D: OBE Academic Workflow	Complete lifecycle: PEO -> PLO -> CLO -> Assessment -> Marks -> Attainment -> CQI.
Part E: Outcome Matrix Reference	PEO-to-PLO alignment, CLO-to-PLO curriculum matrices, and correlation strength scales.
Part F: Mathematical Calculations & RBRI	Exact attainment formulas, RBRI risk index derivation, direct/indirect weighting, and attendance.
Part G: Visualizations & Analytics	Predictive confusion matrix, attendance scatter plots, attainment heatmaps, and SAR dossiers.
Part H: AI/ML Integration Blueprint	Gap analysis of current system, proposed Python FastAPI service, and phased AI roadmap.
Part I: Enterprise Security & Auditing	JWT claims, cryptographic hashing, immutable audit trails, and data minimization.
Part J: Traceable Requirements Matrix	Traceable functional, academic, operational, and non-functional requirements (FR-01 to FR-20).
Part K: Developer & Admin Runbook	Operational commands, database administration, backup procedures, and disaster recovery.

Part A: System Overview & Institutional Scope

Higher education engineering and computing institutions operating under the Washington Accord (administered in Pakistan by the Pakistan Engineering Council - PEC and the National Computing Education Accreditation Council - NCEAC) are required to transition from traditional percentage grading to Outcome-Based Education (OBE). Under OBE, student competency is measured against explicitly articulated learning objectives rather than relative ranking.

The system implements a three-tier hierarchical outcome chain: Program Educational Objectives (PEOs) describing career accomplishments 3-5 years post-graduation; Program Learning Outcomes (PLOs) defining 12 standard graduate attributes; and Course Learning Outcomes (CLOs) detailing specific measurable skills achieved in individual course modules.

Role Key	Role Label	Institutional Scope & Permissions
ADMIN	System Administrator / Dean	University-wide authority: institutional academic policy authoring, user provisioning, curriculum structure, session catalogs, and accreditation reporting.
HOD	Head of Department	Departmental management: course allocation, curriculum review, departmental policy overrides, and CQI verification.
TEACHER	Course Instructor /	Course management: CLO definition, assessment questions authoring, marks grading, attendance

Role Key	Role Label	Institutional Scope & Permissions
	Faculty	recording, and remediation logging.
STUDENT	Undergraduate Student	Personal student portal: individual CLO mastery, PLO competency radar, assessment breakdown, and attendance exam clearance.

Part B: Codebase Architecture & State Management

The frontend is organized into modular React 19 functional components powered by Tailwind CSS v4. Centralized state management is implemented in src/services/dataStore.ts using an observable store pattern that transparently synchronizes with the Express REST API and PostgreSQL database.

All computational intelligence and mathematical formulas reside in a pure, deterministic engine: src/services/attainmentEngine.ts. This separation ensures that mathematical attainment calculations can be executed both within the client browser and headlessly inside Node.js automated audit suites without DOM dependencies.

File / Module Path	Architectural Role & Functional Responsibility
src/types.ts	Core TypeScript domain models: AcademicPolicy, PEO, PLO, CLO, Assessment, Mark, Attendance, CQI.
src/services/attainmentEngine.ts	Pure calculation engine: calculateClassCLOAttainment, calculatePLOAttainment, resolveActivePolicy, evaluateAttendanceEligibility.
src/services/dataStore.ts	Observable application store handling optimistic updates, API mutations, and reactive event notification.
src/services/exportService.ts	Official PDF and Excel export engine utilizing jsPDF and xlsx for SAR and transcript packages.
src/components/OutcomesHierarchyModule.tsx	Interactive PEO-to-PLO and CLO-to-PLO mapping matrix with Level 1-3 correlation strength scales.
src/components/AttendanceModule.tsx	Independent attendance register with statutory examination debarment and review workflow flags.
src/components/AdminAnalyticsDashboard.tsx	Comprehensive analytics suite featuring confusion matrix, scatter correlations, and CQI trackers.
server.ts	Express API server hosting REST endpoints, bcrypt authentication, and mounting Vite development middleware.

Part C: Database Design, Schema & Constraints

The database model adheres to strict Boyce-Codd Normal Form (BCNF) with relational integrity enforced through foreign key constraints with ON DELETE CASCADE. Primary keys use cryptographically unique strings or UUIDs.

Table Name	Primary Key	Foreign Keys	Description & Business Rules
departments	id TEXT	None	Academic faculties / departments (e.g. Computing, Electrical).
programs	id TEXT	department_id	Degree programs (e.g. BSCS, BSSE) with accreditation body.
academic_sessions	id TEXT	None	Academic calendar terms (e.g. Fall 2025, Spring 2026).
courses	id TEXT	department_id, program_id	Course catalog with official credit hours (e.g. CS-301).
course_offerings	id TEXT	course_id, session_id, teacher_id	Active semester class sections assigned to faculty.
enrollments	id TEXT	offering_id, student_id	Student class enrollments with registration status.
peos	id TEXT	program_id	Program Educational Objectives with career benchmark %.
plos	id TEXT	program_id	Program Learning Outcomes (Washington Accord Graduate Attributes).
clos	id TEXT	course_id, plo_id	Course Learning Outcomes with Bloom taxonomy cognitive level.
peo_plo_mappings	id TEXT	peo_id, plo_id	Curriculum mapping table storing correlation weights (0.1 to 1.0).
assessments	id TEXT	offering_id	Exams, quizzes, assignments with weightage and max marks.
questions	id TEXT	assessment_id, clo_id	Granular question items mapped to specific CLOs.

Table Name	Primary Key	Foreign Keys	Description & Business Rules
marks	id TEXT	question_id, student_id	Obtained scores with check constraint (0 <= obtained <= max).
attendance_records	id TEXT	offering_id, student_id	Session attendance entries (Present, Absent, Late, Excused).
academic_policies	id TEXT	dept_id, prog_id, course_id	Configurable institutional thresholds and aggregation rules.
cqi_actions	id TEXT	course_id, outcome_code	Continuous Quality Improvement closed-loop remediation actions.

Part D: Complete OBE Academic Workflow Lifecycle

The OBE operational workflow operates in eight sequential phases, transforming institutional objectives into verifiable classroom learning evidence:

Phase 1: Institutional Objective Definition — HODs and academic committees define PEOs (career goals) and PLOs (graduate attributes) for each degree program.

Phase 2: Curriculum Mapping & Alignment — Faculty map PEOs to PLOs with weighted correlation factors (Strong, Moderate, Low), and map Course CLOs to PLOs.

Phase 3: Assessment Authoring & Question Blueprinting — Instructors create midterms, finals, and lab projects, mapping each specific question to an intended CLO and Bloom cognitive level (e.g. Q1 -> CLO-1 C4).

Phase 4: Marks Grading & Rubric Evaluation — Student marks are recorded question-by-question. The system enforces strict boundary validation (0 <= marks <= maxMarks).

Phase 5: CLO Attainment Computation — For each CLO, the engine evaluates individual student percentage against the pass threshold (e.g. 50%), and calculates the cohort percentage meeting the criterion.

Phase 6: Direct & Indirect PLO Rollup — CLO achievements roll up into direct PLO scores. Course exit surveys provide indirect assessment scores. These are combined using the configured institutional policy weights (e.g. 80% Direct + 20% Indirect).

Phase 7: Independent Attendance Clearance — The attendance engine independently evaluates lecture presence. Students falling below the statutory threshold (e.g. 75%) are marked BARRED / INELIGIBLE for examination.

Phase 8: Closed-Loop Continuous Quality Improvement (CQI) — Any outcome falling below the benchmark triggers a mandatory CQI action. Faculty document root cause, proposed remediation, and verify effectiveness in the subsequent offering.

Part E: Outcome Matrix Reference & Correlation Standards

The system implements standard three-tier correlation scaling for curriculum matrices in accordance with PEC and Washington Accord standards:

Correlation Scale	Numeric Weight	Educational Significance	Curricular Role
Strong (Level 3 / S)	1.00 (or 3)	Substantial / High Alignment	Core focus of the objective; extensive instruction and major assessment weight.
Moderate (Level 2 / M)	0.66 (or 2)	Medium Alignment	Significant contributing component; moderate classroom practice and secondary assessment.
Low (Level 1 / L)	0.33 (or 1)	Introductory / Peripheral Alignment	Foundational prerequisite or introductory exposure; minimal direct grading.
None (—)	0.00 (or 0)	No Curricular Alignment	Outcome is not covered in this objective or course module.

Part F: Mathematical Formulations & RBRI Metric

1. Course Learning Outcome (CLO) Attainment Formula

Individual Student CLO Score (%): For a given student and CLO, the score is the ratio of obtained marks to maximum marks across all questions mapped to that CLO:
Score_CLO = (Sum(obtainedMarks_q) / Sum(maxMarks_q)) * 100
A student achieves the CLO if Score_CLO >= studentPassThresholdPct (Configured default: 50%).

Cohort CLO Attainment Percentage (%): The proportion of assessed enrolled students who met the individual pass criterion:
Attainment_CLO = (Count(Students Meeting Pass Criterion) / Count(Eligible Assessed Students)) * 100
The cohort meets the CLO if Attainment_CLO >= cohortKpiTargetPct (Configured default: 50%). Note that this percentage is mathematically distinct from the class average score!

2. Program Learning Outcome (PLO) Direct, Indirect & Combined Attainment

Combined PLO Attainment Formula:
A_combined = w_direct * A_direct + w_indirect * A_indirect
where w_direct + w_indirect = 1.0 (or 100%). In accordance with PEC/HEC standards, weights default to 80% direct and 20% indirect, or can be configured to 70/30 or other statutory ratios.

3. The RBRI Metric: Risk-Based Remediation Index

In this codebase, RBRI (Risk-Based Remediation Index / Risk-Based Retention Index) represents a multi-factor diagnostic indicator computed for every enrolled student. It synthesizes academic marks performance, lecture attendance consistency, and failed prerequisite CLO counts to classify students into actionable risk tiers (HIGH RISK, MEDIUM RISK, LOW RISK).

RBRI Mathematical Derivation:
RBRI_Score = 0.45 * (100 - AvgMarks) + 0.35 * (100 - AttendancePct) + 0.20 * (FailedCLOCount * 25)
Where:
- RBRI >= 50 or Attendance < 75%: Flagged as HIGH RISK (debared from exam / urgent remedial advising)
- 30 <= RBRI < 50: Flagged as MEDIUM RISK (watchlist / supplemental tutoring)
- RBRI < 30: Flagged as LOW RISK (satisfactory good standing)

4. Independent Attendance Eligibility Formula

Effective Attendance Rate (%):
EffectiveDenominator = Max(1, TotalScheduledSessions - ExcusedSessions)
AttendancePct = Round((AttendedSessions / EffectiveDenominator) * 100)
Status is evaluated as:
- MEETS_REQUIREMENT: AttendancePct >= Threshold (Eligible for Final Exam)
- BELOW_REQUIREMENT: AttendancePct < Threshold and 0 Excused (Debarred from Final Exam)
- REQUIRES_AUTHORIZED_REVIEW: AttendancePct < Threshold but ExcusedSessions > 0 (Referred to HoD Committee)
- PENDING_EVALUATION: TotalSessions == 0 (Incomplete data, never assumed to be 0%)

Part G: Visualizations & Analytics Suite

The application features a comprehensive visualization suite designed to communicate actionable intelligence to institutional stakeholders:

Visualization Type	Analytical Purpose & Educational Interpretation
Predictive Confusion Matrix	Evaluates predictive model accuracy for identifying at-risk students. Displays True Positives (TP), False Positives (FP), True Negatives (TN), False Negatives (FN), along with Precision, Sensitivity (Recall), and Accuracy metrics.
Attendance vs. Marks Scatter Plot	Explores correlation between session presence percentage and examination marks, illustrating the empirical relationship between attendance and academic achievement.
PLO Attainment Grouped Bar Chart	Displays Direct, Indirect, and Combined attainment bars for each of the 12 Washington Accord PLOs against the institutional benchmark line.
Curriculum Mapping Matrix	Color-coded matrix displaying PEO-to-PLO correlation strengths with interactive level cycling and coverage gap indicators.
Student Risk Donut Distribution	Segmented breakdown of cohort students in Good Standing, Watchlist, and Examination Ineligible status.
Accreditation SAR Package PDF	Complete Self-Assessment Report generated on demand for external accreditation review teams.
Student OBE Transcript PDF	Individual certified transcript showing granular CLO mastery and PLO graduate attribute competencies.

Part H: AI/ML Integration Blueprint & Architectural Roadmap

Current Application State & Gap Analysis: The current production codebase does not include an active machine learning inference engine. Statistical heuristics, deficiency scanning, and rule-based confusion matrix calculations are currently performed deterministically in TypeScript. Machine learning integration represents a planned future architectural enhancement.

Proposed Microservices Architecture: We recommend deploying a lightweight Python FastAPI microservice alongside the Node.js/Express backend. This service will ingest sanitized assessment data from PostgreSQL and expose inference endpoints for NLP course document parsing and predictive risk trajectory modeling.

Prioritized AI/ML Feature	Recommended ML Technique	Educational Value	Human-in-the-Loop Safety
Automated Deficiency Detection	Time-series forecasting / Anomaly detection (Isolation Forest)	Detects declining outcome trends across semesters before official accreditation audits.	Advisory only. Results presented to QEC for statutory review.
AI-Assisted CQI Recommendations	Fine-tuned LLM (Gemini 2.5 Flash / Llama 3) with RAG	Drafts specific pedagogical interventions based on failed question topics and Bloom levels.	Mandatory HoD / Instructor approval required before CQI action is recorded.
NLP Course Document Parsing	Named Entity Recognition (NER) & Sentence Transformers	Extracts proposed CLOs, Bloom verbs, and question alignments from course syllabus PDFs.	Instructor must review and accept proposed mappings in curriculum manager.
Student Early Warning Trajectory	Gradient Boosted Decision Trees (XGBoost / LightGBM)	Identifies students on trajectory for academic probation prior to midterm exams.	Protected against algorithmic bias; never exposes sensitive demographic data.
Accreditation Audit RAG Assistant	Retrieval-Augmented Generation over SAR and institutional docs	Instantly answers PEC accreditation evaluator inquiries with citations to stored evidence.	Directly links evaluator to verified student marksheets and exam papers.

Part J: Traceable Requirements Matrix (FR-01 to FR-20)

Req ID	Requirement Description	Target Role	Current Status	Verification Method
FR-01	Multi-role authentication with JWT tokens and bcrypt hashing	ALL	VERIFIED & COMPLETE	Automated login test in tests/verify-obe-workflow.ts.
FR-02	Configurable institutional academic policies with 80/20 weights	ADMIN	VERIFIED & COMPLETE	Unit tests validating policyMissing and custom weights.
FR-03	Hierarchical policy scoping (Course -> Program -> Dept -> University)	ADMIN	VERIFIED & COMPLETE	Hierarchical precedence engine test Step 13.
FR-04	Course catalog, credit hours preservation, and offering allocation	ADMIN	VERIFIED & COMPLETE	Integration test Step 2-5 verifying 4 credits preserved.
FR-05	PEO, PLO, and CLO authoring with Bloom taxonomy levels	ADMIN / TEACHER	VERIFIED & COMPLETE	Curriculum management tests Step 6.
FR-06	Weighted correlation matrices (PEO-PLO and CLO-PLO)	ADMIN / TEACHER	VERIFIED & COMPLETE	Interactive matrix UI with Strong, Moderate, Low scales.
FR-07	Question-by-question marks entry with boundary enforcement	TEACHER	VERIFIED & COMPLETE	Boundary validation test rejecting -5 and 30/25.
FR-08	Deterministic CLO attainment calculation (students achieved/eligible)	TEACHER / QEC	VERIFIED & COMPLETE	Step 9 verifying Alice 80% and Bob 32% precision.
FR-09	Direct, Indirect, and Combined PLO attainment aggregation	QEC / DEAN	VERIFIED & COMPLETE	PLO direct/indirect weighting test Step 9 & 12.
FR-10	Closed-loop Continuous Quality Improvement (CQI) logging	TEACHER / HOD	VERIFIED & COMPLETE	Deficiency scanner detecting NEEDS_CQI flag Step 10.
FR-11	Independent attendance examination eligibility calculator	ALL	VERIFIED & COMPLETE	Step 14 verifying Bob debarred at 50% vs 75% rule.
FR-12	Predictive confusion matrix with Precision, Recall, and Accuracy	ADMIN	VERIFIED & COMPLETE	Category 3 analytics rendering in Admin Dashboard.
FR-13	Official Washington Accord SAR / COAP PDF generation	ADMIN / QEC	VERIFIED & COMPLETE	Tested exportService.generateAccreditationSARPDF.
FR-14	Individual Student certified OBE transcript PDF export	STUDENT / ADMIN	VERIFIED & COMPLETE	Tested exportService.generateStudentOBETranscriptPDF.
FR-15	Row-Level Security (RLS) enforcement on all Postgres tables	ALL	VERIFIED & COMPLETE	Supabase migrations 001 and 003 RLS policies.
FR-16	AI-assisted CQI recommendation generation via LLM	TEACHER	PLANNED / PHASE 2	Documented in AI/ML Integration Blueprint Part H.
FR-17	NLP syllabus document parsing for candidate CLO extraction	TEACHER	PLANNED / PHASE 2	Documented in AI/ML Integration Blueprint Part H.
FR-18	Student failure risk predictive model with gradient boosting	ADVISOR	PLANNED / PHASE 3	Documented in AI/ML Integration Blueprint Part H.
FR-19	RAG accreditation audit chatbot with verified citations	EVALUATOR	PLANNED / PHASE 3	Documented in AI/ML Integration Blueprint Part H.
FR-20	Multi-campus institutional benchmarking federation	SUPER_ADMIN	PROPOSED /	Roadmap item for inter-university accreditation

Req ID	Requirement Description	Target Role	Current Status	Verification Method
			FUTURE	consortium.